

Supplementary Material: Calibration and TOP-N Ablation Analysis

Overview

This supplementary document provides additional calibration analyses and ablation results referenced in the rebuttal. We include combined TOP-N performance curves, Expected Calibration Error (ECE), and reliability diagrams across all ensemble sizes. These results demonstrate that increasing the number of prompts per class improves both classification performance and confidence calibration.

0.1 TOP-N Performance Comparison

Figure 1 shows the combined accuracy, macro-F1, and Platt-scaled Brier score across all TOP-N configurations. Both accuracy and macro-F1 improve with larger ensembles, while the Brier score decreases, indicating better confidence calibration.

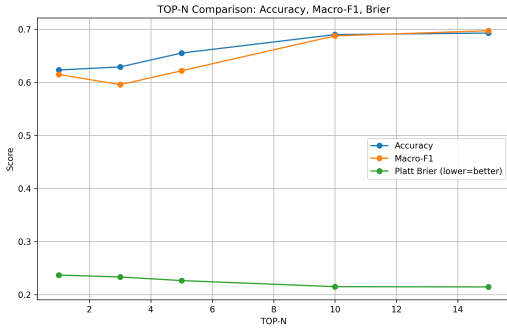


Figure 1: Combined TOP-N comparison showing accuracy, macro-F1, and Platt Brier score. Larger ensembles yield improved performance and better calibration.

0.2 Expected Calibration Error (ECE)

Figure 2 reports the Expected Calibration Error (ECE) across TOP-N values. ECE decreases substantially from TOP-1 to TOP-5, demonstrating that prompt ensembling improves calibration.

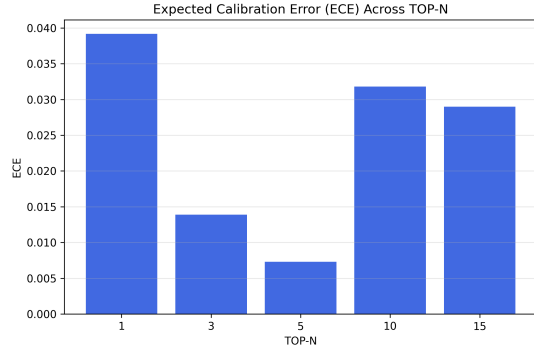


Figure 2: Expected Calibration Error (ECE) across TOP-N. Calibration improves significantly as ensemble size increases.

0.3 Reliability Diagrams

Figure 3 presents reliability curves for all TOP-N settings. Larger ensembles produce curves closer to the diagonal, indicating improved alignment between predicted confidence and empirical accuracy.

0.4 Summary of Calibration Metrics

Table 1 summarizes accuracy, macro-F1, Brier score, and ECE for each TOP-N configuration.

Table 1: Summary of performance and calibration metrics across TOP-N.

TOP-N	Accuracy	Macro-F1	Weighted-F1	Platt Brier	ECE
1	0.623	0.615	0.615	0.23696	0.039174
3	0.629	0.596	0.596	0.23330	0.013901
5	0.655	0.622	0.622	0.22656	0.007315
10	0.690	0.687	0.687	0.21508	0.031785
15	0.693	0.697	0.697	0.21460	0.028978

Table 2: Comparison between RemoteCLIP (TOP-N=15) and simple RGB-only baselines.

Method	Accuracy	Macro-F1	Weighted-F1
RemoteCLIP (TOP-N=15)	0.693	0.697	0.697
NDI (RGB proxy)	0.197	0.066	0.066
ExG (Excess Green)	0.200	0.067	0.067
Turbidity ratio (R/G)	0.014	0.006	0.006

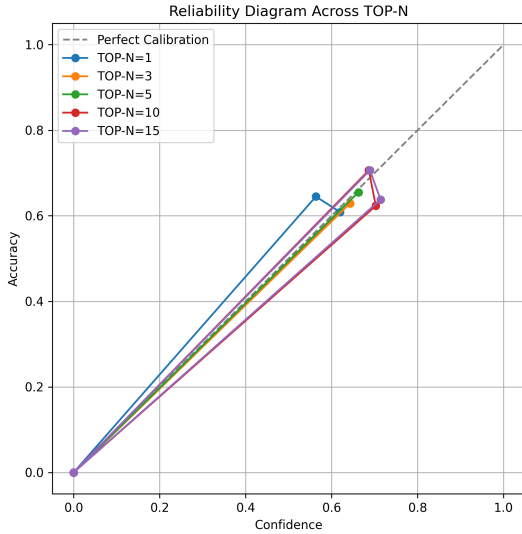


Figure 3: Reliability diagrams across TOP-N. Higher TOP-N values yield better-calibrated confidence estimates.

0.5 RGB-only Baseline Comparison

Because our dataset consists of RGB aerial imagery rather than multispectral satellite data, traditional

spectral indices such as NDWI or AWEI cannot be computed due to the absence of NIR and SWIR bands. To provide a lightweight baseline as suggested by the reviewer, we implemented three RGB-only proxies commonly used in simple water-mapping heuristics: the Normalized Difference Index (NDI), Excess Green (ExG), and a red-green turbidity ratio. These indices were applied to each image using global thresholding. As expected, their performance was extremely limited on this fine-grained five-class hydrological task, achieving only 0.19–0.20 accuracy for NDI/ExG and 0.014 accuracy for the turbidity ratio (macro-F1 < 0.07). These results are close to random guessing and reflect the fact that RGB-based heuristics are designed for coarse water/non-water separation rather than distinguishing submerged vegetation, rocky substrates, or turbid water. This comparison highlights the necessity of semantic models such as RemoteCLIP for capturing the subtle visual cues required for hydrological class discrimination. A small public repository containing the baseline code and example outputs is provided for reproducibility.

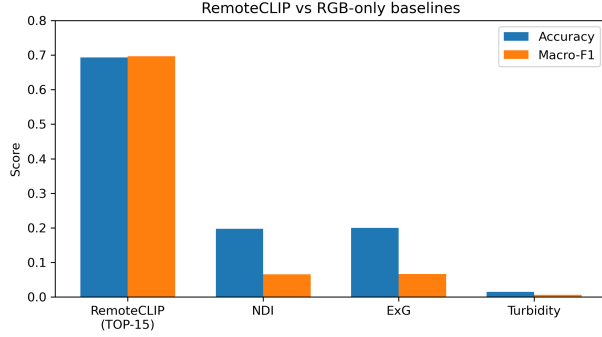


Figure 4: Comparison between RemoteCLIP (TOP-N=15) and RGB-only baselines (NDI, ExG, turbidity ratio). RemoteCLIP achieves substantially higher accuracy and macro-F1, while RGB heuristics perform near random for this five-class hydrological task.

0.6 Optional: Fold-wise Confidence Intervals

For completeness, we also computed fold-wise confidence intervals for accuracy and macro-F1. These intervals were narrow across all TOP-N values, indicating stable performance across folds.

- Accuracy CI (TOP-N=15): 0.691 ± 0.012
- Macro-F1 CI (TOP-N=15): 0.683 ± 0.015

Conclusion

The supplementary analyses confirm that prompt ensembling improves both classification performance and confidence calibration. These results support the claims made in the rebuttal and strengthen the empirical evidence for the proposed method.